

APAVS



Academic Title

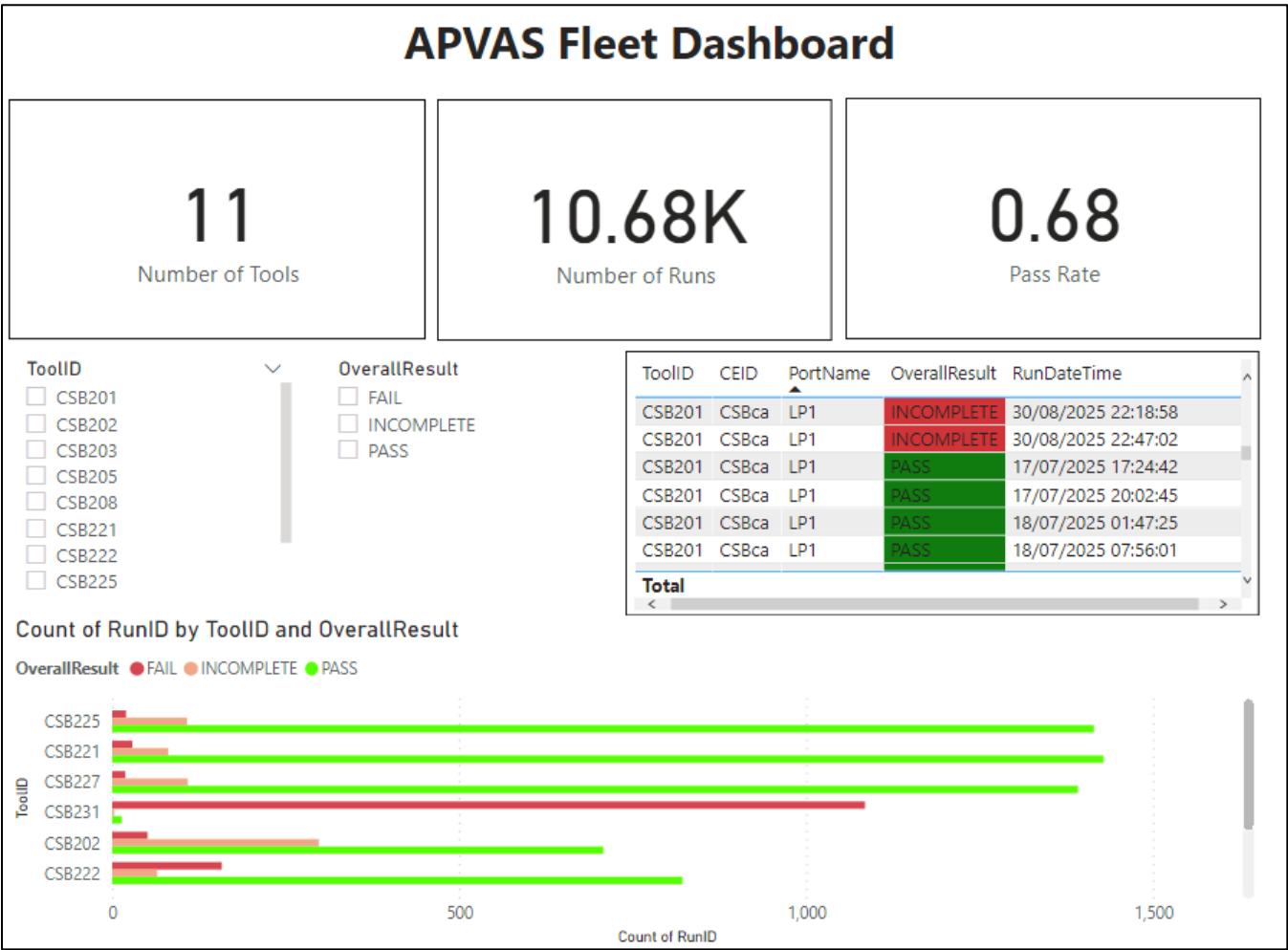
Automated Pipeline and Visualisation System for Tool Load Port Data

Project Areas

- Data Analytics
- Automation
- Semiconductor Manufacturing

Project Supervisor: Sonya Hogan

Abstract - APAVS automates the monitoring of load port qualification data across a fleet of semiconductor manufacturing tools. Currently, technicians manually export log files, run Excel macros, and visually inspect qualification plots for individual tools with no fleet-wide visibility. APAVS replaces this workflow with a PowerShell extraction pipeline that retrieves data directly from tool hard drives, parses raw sensor logs, and calculates slot-level offsets against taught reference positions. Results are stored in a SQL Server database and visualised through a Power BI dashboard providing real-time pass/fail status, trend analysis, and fleet-wide visibility, enabling engineers to identify drift before it impacts production.



Technologies: PowerShell, SQL Server, Power BI

URL: <https://bit.ly/APAVS->

By Vincent Byrne